1. a. A random variable takes values X=0 with probability $\frac{24}{25}$ and the value X=1 with probability $\frac{1}{25}$. Find the bound-on probability that X is at least 5. (2)

Solution:

As the random variable X takes values X=0 with probability $\frac{24}{25}$ and the value X=1 with probability $\frac{1}{25}$, the expectation of X is $E(X) = 0\times\frac{24}{25} + 1\times\frac{1}{25} = \frac{1}{25}$.

Therefore, $\Pr ob(X \geq 5) \leq \frac{E(X)}{5} = \frac{\frac{1}{25}}{5} = \frac{1}{125}$.

b. If X is the number scored in a throw of a fair die, show that the Chebyshev's inequality gives $\text{prob}\{|X-\mu| > 2.5\} < 0.47.$ while the actual probability is zero. (6)

Solution:

Here $E(X) = \frac{1}{6}\times(1+2+3+4+5+6) = \frac{7}{2}$ and $E(X^2) = \frac{1}{6}\times(1^2+2^2+3^2+4^2+5^2+6^2) = \frac{91}{6}$

So $\sigma^2 = \frac{35}{12} = 2.9167$

By Chebyshev's inequality

$$\text{prob}\{|X-\mu| > k\} \leq \frac{\text{Var}(X)}{k^2}$$

$$\Rightarrow \text{prob}\{|X-\mu| > 2.5\} < \frac{2.9167}{6.25} = 0.47$$

The actual probability is p= $\text{prob}\{|X-3.5| > 2.5\}$

$$= \Pr ob\left(X \text{ lies outside the limits } (3.5 - 2.5.\ 3.5 + 2.5) = (1,6)\right)$$

As X is the event occurred when the dice is thrown, X cannot lie outside (1,6).

Hence, actual probability is 0.